

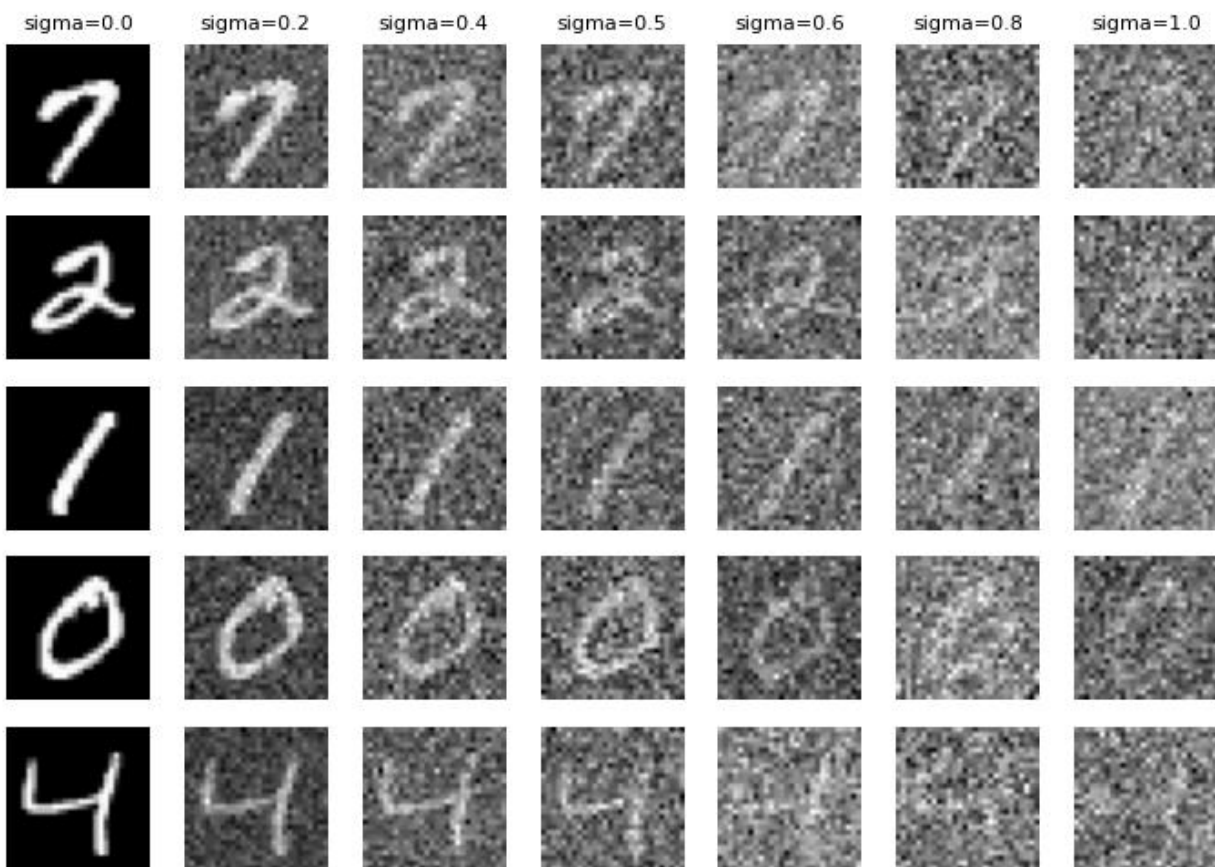
Assignment 5: Diffusion Models

Name(s): Licheng Xu & Libin Wang

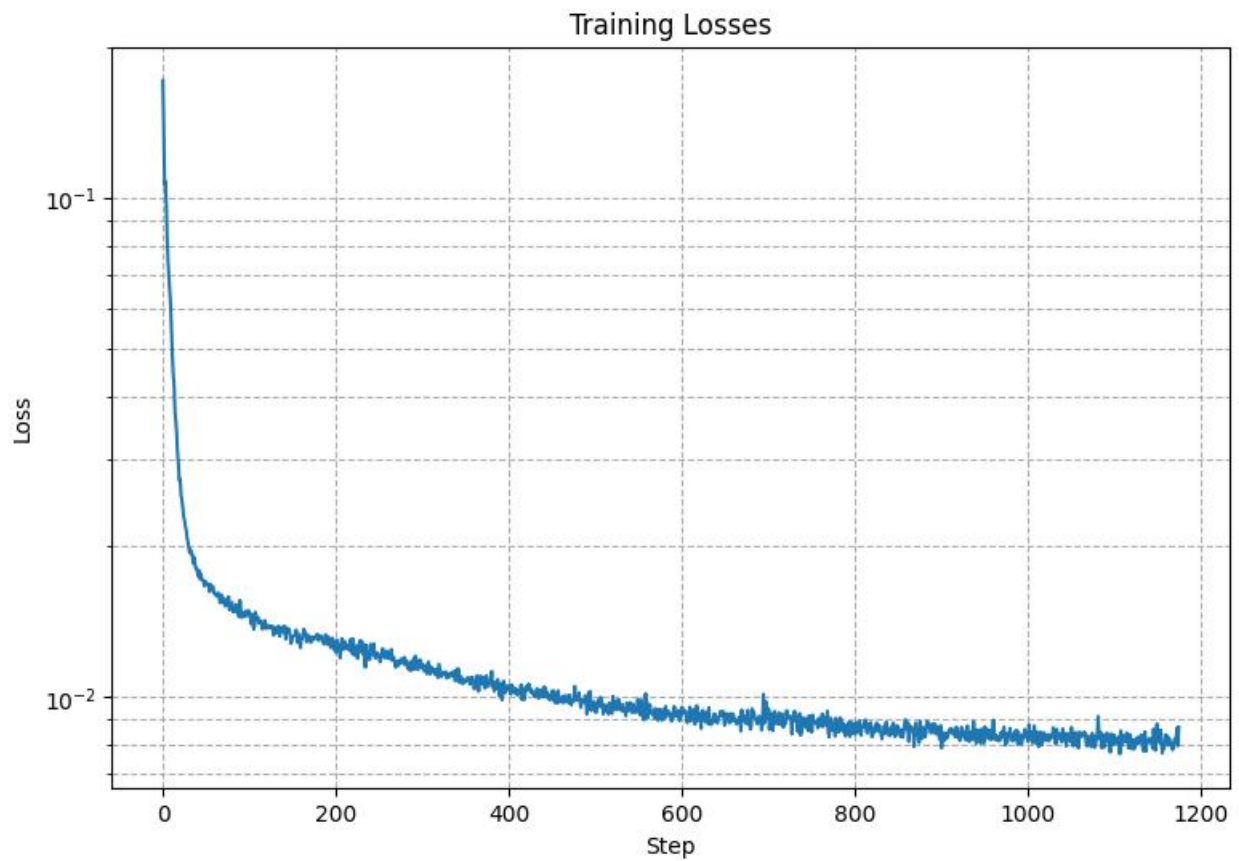
NetID(s): lx8 & libin2

Part 1: Single-Step Denoising UNet

1. A visualization of the noising process using $\sigma = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0]$ (Figure 3)

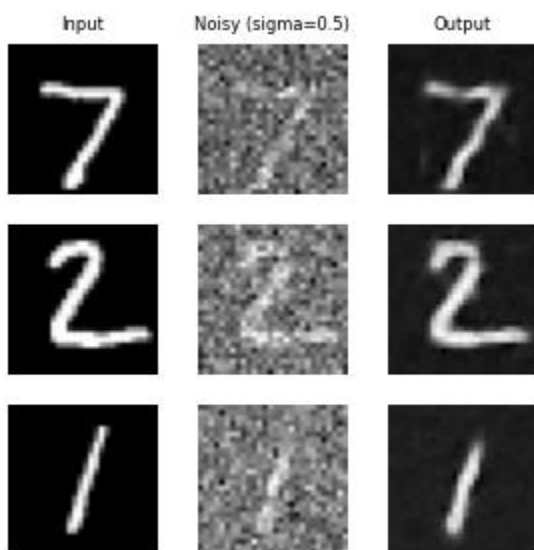


2. A training loss curve plot every few iterations during the whole training process (Figure 4)

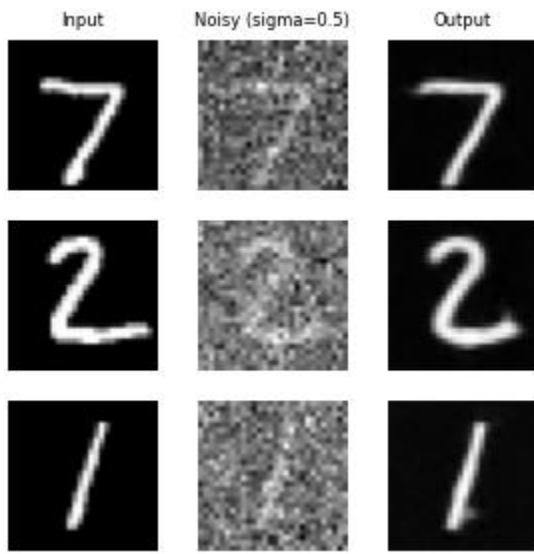


3. Sample results on the test set after the first and the 5th epoch. (Figures 5 and 6)

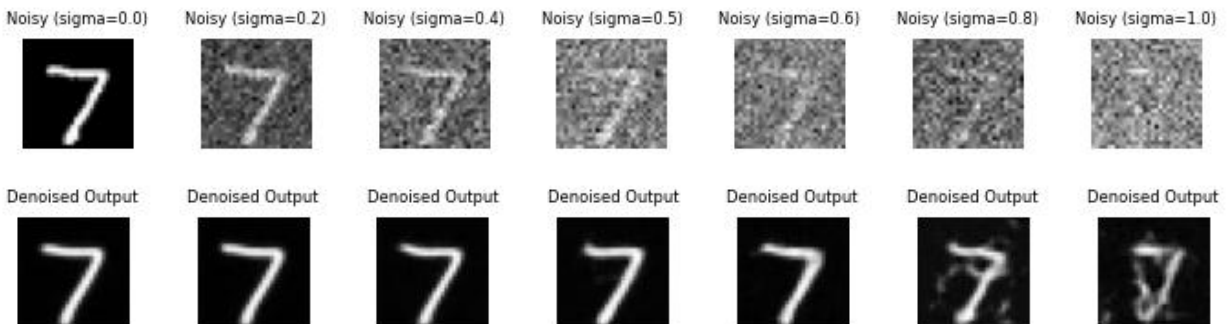
The first epoch:



The 5th epoch:



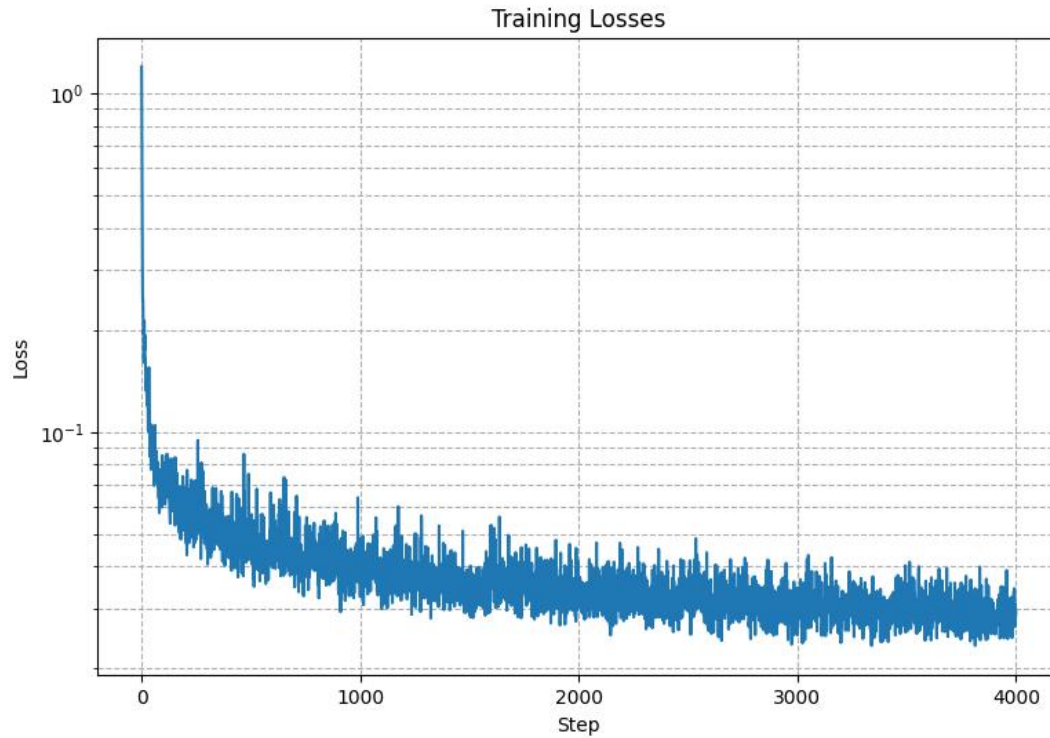
4. Sample results on the test set with out-of-distribution noise levels after the model is trained. Keep the same image and vary $\sigma = [0.0, 0.2, 0.4, 0.5, 0.6, 0.8, 1.0]$ (Figure 7)



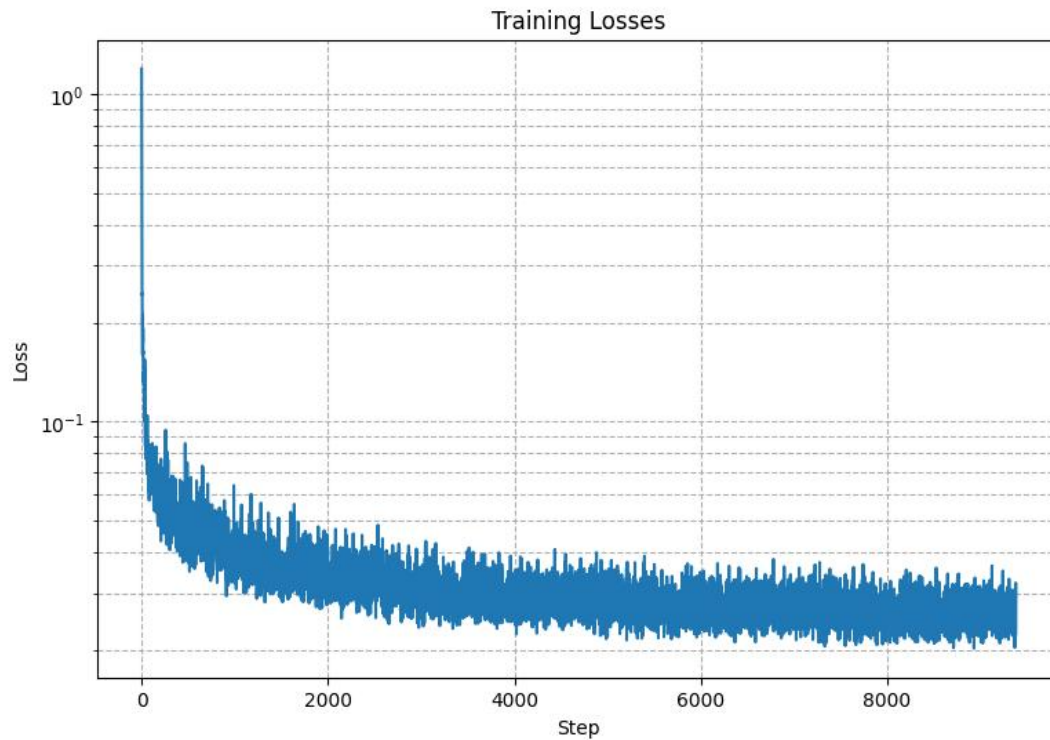
Part 2.1-2.3: Time-conditioned UNet

1. A training loss curve plot for the time-conditioned UNet over the whole training process (Figure 10).

The training loss curve for first 4000 steps:



The training loss curve for whole training process:



2. Sampling results for the class-conditioned UNet after the 5th and the 20th epoch. Providing a GIF is optional and can be done as extra credit.

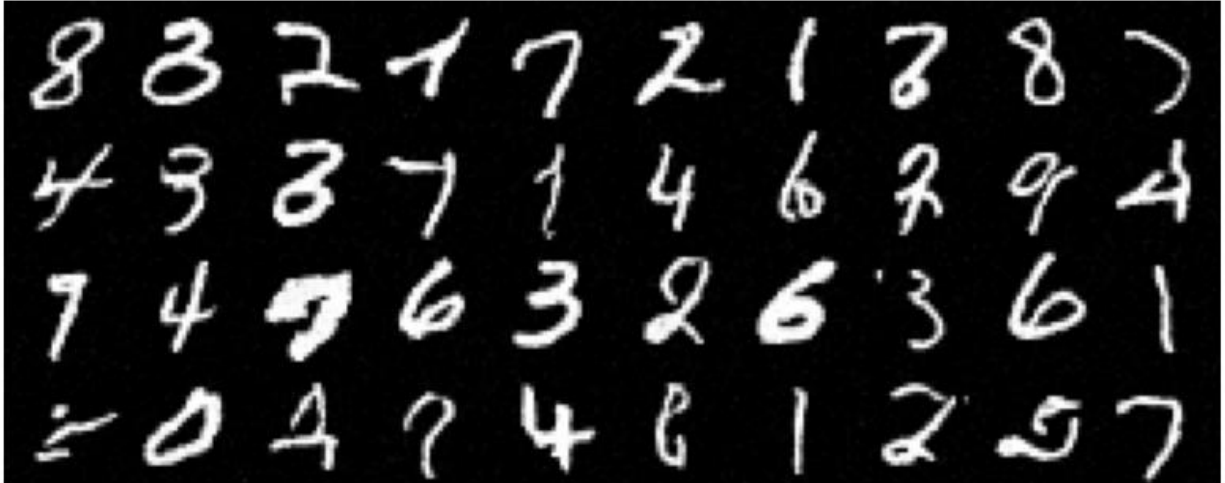
The 5th epoch:

Samples at Epoch 5



The 20th epoch:

Samples at Epoch 20

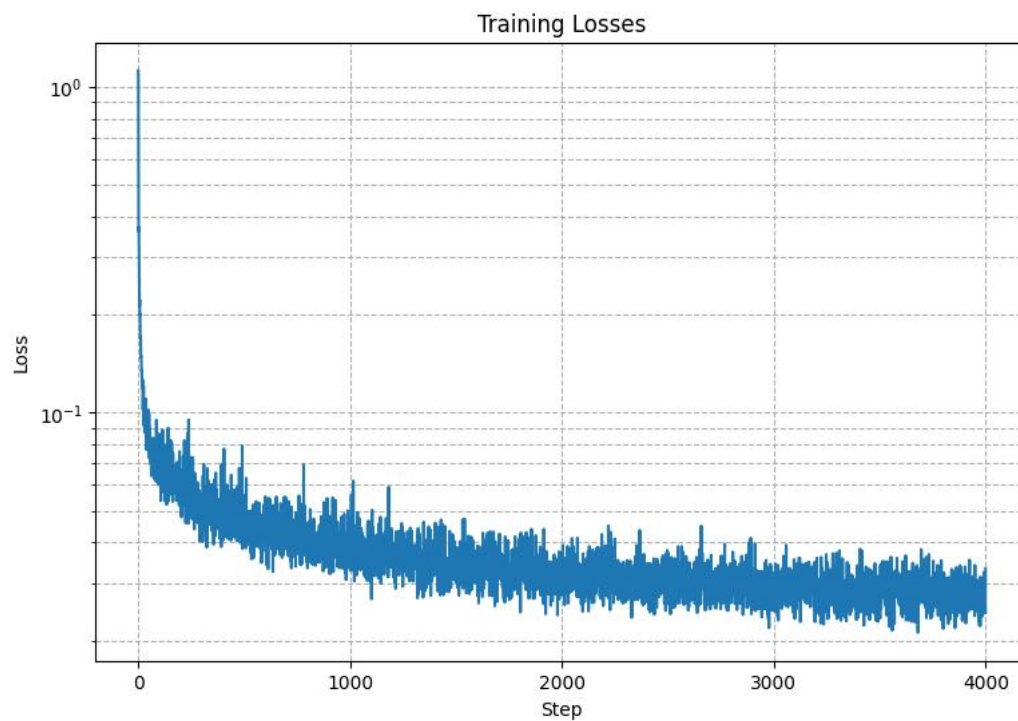


(We generate GIFs for 1st, 5th, 10th, 15th and 20th epoch, included in the .zip file)

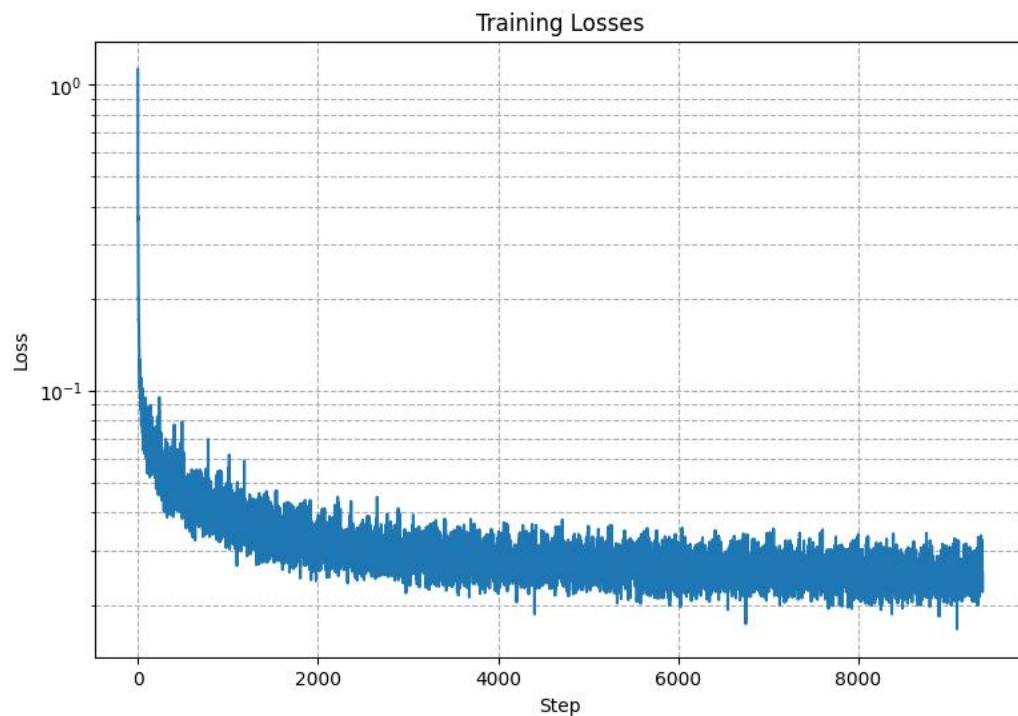
Part 2.4-2.5: Class-conditioned UNet

1. A training loss curve plot for the class-conditioned UNet over the whole training process.

The training loss curve for first 4000 steps:



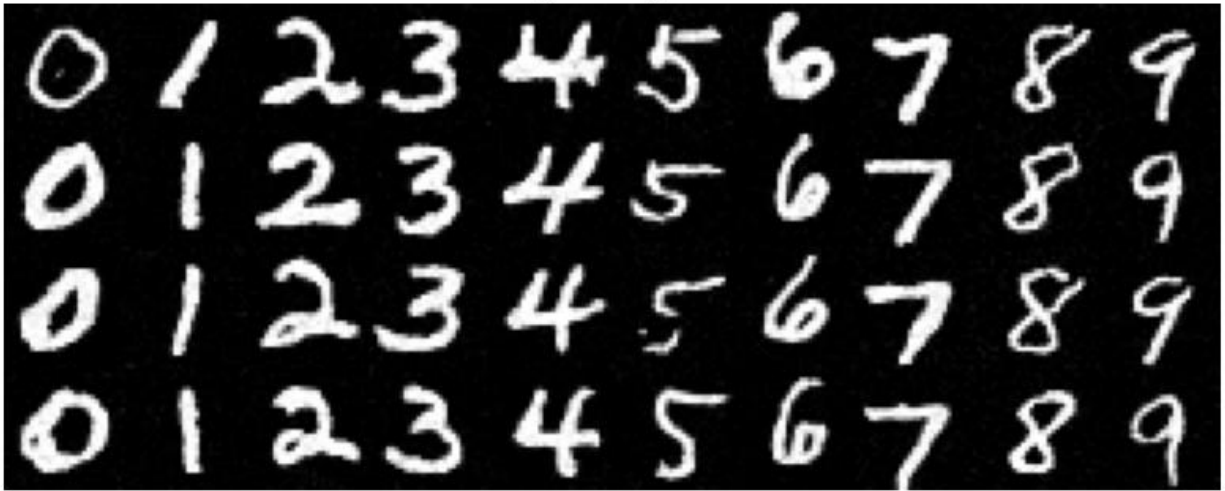
The training loss curve for whole training process:



2. **Sampling results for the class-conditioned UNet after the 5th and the 20th epoch. Generate 4 instances of each digit as shown on the assignment page. Providing a GIF is optional and can be done as extra credit.**

The 5th epoch:

Samples at Epoch 5



The 20th epoch:

Samples at Epoch 20



(We generate GIFs for 1st, 5th, 10th, 15th and 20th epoch, included in the .zip file)

Extra credit:

We show the process of DDPM and Class-conditioned DDPM, from pure noise to the generated digits images, in .gif format. We choose one time step out of every 10 as a frame, and are generating the corresponding gif images for 1st, 5th, 10th, 15th and 20th epoch. The files are included in the .zip file.